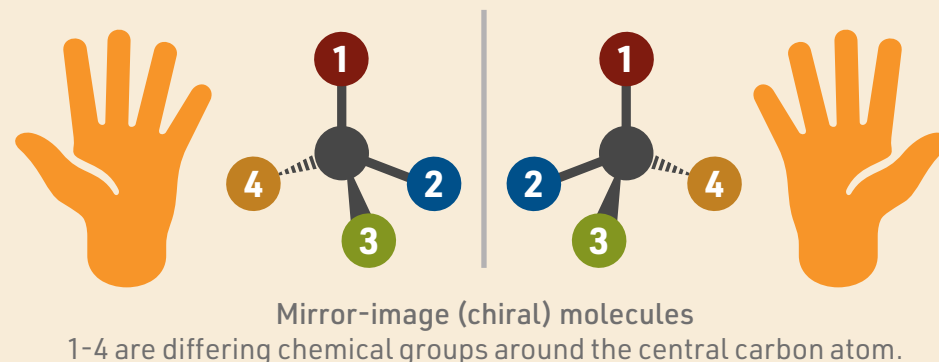


The 2026 Nobel Prize in Chemistry

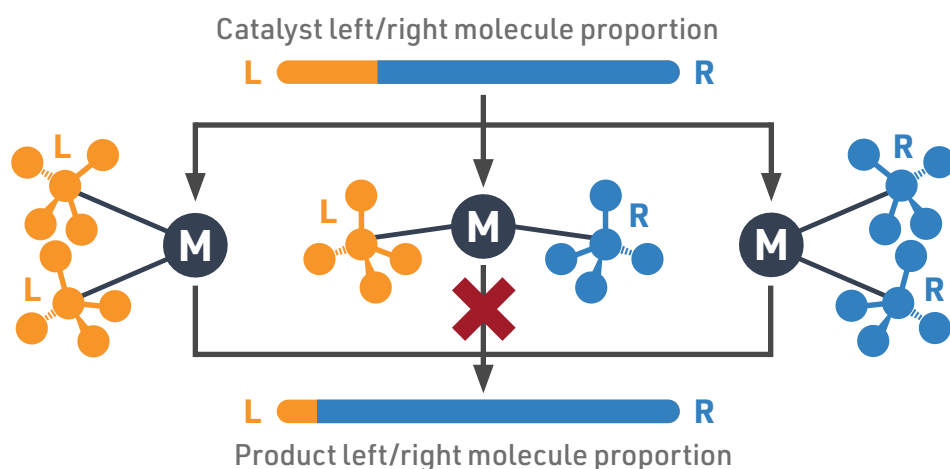


The 2026 Nobel Prize in Chemistry was awarded to **Henri Kagan** and **Kenso Soai** for the discovery of non-linear effects and autocatalysis in asymmetric organic synthesis.

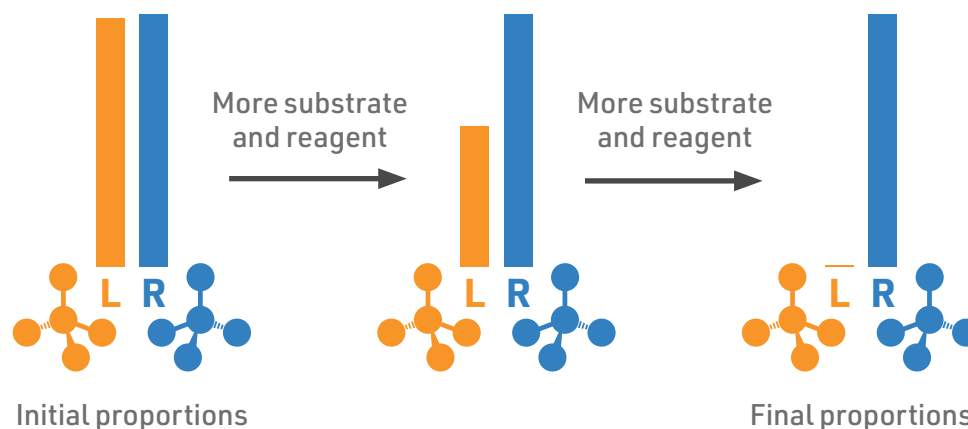
Many carbon-based molecules are chiral – they have mirror image forms, much like our hands. Almost all proteins in living organisms are built from left-handed versions of amino acids. However, lab reactions often produce both mirror images of molecules. This year's Nobel Prize goes to research on reactions that help chemists produce only one mirror image of a molecule.



In 1986, **Henri Kagan** found a way to boost the formation of one of the chiral products in chemical reactions. The catalyst for these reactions consists of a metal ion attached to a different chiral molecule. Kagan discovered that by attaching a mix of left and right molecules to the metal ion, he could produce *non-linear effects* – that is, proportions of left and right product molecules differed from the proportion of the left and right molecules used with the catalyst.



In 1995, **Kenso Soai** designed the first ever lab-based chemical reaction which made only one of the product's possible mirror images. He reacted a non-mirror-image molecule to form a chiral product. Once formed, this product speeds up the creation of more of the same product – *autocatalysis*. By starting with a small excess of either the left or right molecules, this autocatalysis can eventually lead to the excess molecule forming 99.99% of the reaction product.



WHY DOES THIS RESEARCH MATTER?

The different mirror images of some molecules can have different properties. This can be of great importance for the synthesis of medicines, as well as providing insights into why biological molecules are formed from single mirror image building blocks.

Nobel Prize in Chemistry press release: <https://www.nobelprize.org/prizes/chemistry/2026/press-release/>